Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 1

September 11, 2026

Exercise 1. Determine the truth values (i.e., T or F) of the following propositions:

- ☐ $19 - 4 = 12$ if and only if 3 is a prime number.
- ☐ If $1 + 1 = 5$, then $1 + 1 = 3$.
- ☐ If the moon is a star, then so is the sun.
- ☐ If 5 is a prime number, then the earth is flat.
- ☐ $0 > 1$ if and only if $2 > 1$.
- ☐ Either Toronto is the capital of Canada or Hamburg is the capital of Germany.

Exercise 2. Construct a truth table for each of these compound propositions:

1. $p \oplus (p \vee q)$
2. $p \wedge (q \oplus u)$
3. $(p \wedge q) \oplus (p \wedge u)$
4. $(p \leftrightarrow q) \oplus (p \rightarrow q)$

Exercise 3. Without using truth tables, show the following logical equivalences:

1. $\neg p \leftrightarrow q \equiv p \leftrightarrow \neg q$
2. $p \oplus (q \wedge u) \not\equiv (p \oplus q) \wedge (p \oplus u)$.
3. $p \oplus q \equiv (p \vee q) \wedge (\neg p \vee \neg q)$.
4. $\neg(p \oplus q) \equiv (\neg p) \oplus q$.
5. $p \leftrightarrow q \equiv \neg(p \oplus q)$.

Exercise 4. Find a compound proposition with three variables p, q, u that is

1. True if and only if p is true, q is false, and u is false;
2. True if and only if exactly one of the variables is true;
3. True if and only if at least two of the variables are true.

Exercise 5. (Rosen, exercise 38, sec. 1.2) Five friends have access to a chat room. Is it possible to determine who is chatting if the following information is available:

- Either Kevin or Heather, or both, are chatting.
- Either Randy or Vijay, but not both, are chatting.
- If Abbey is chatting, so is Randy.
- Vijay and Kevin are either both chatting or neither is.
- If Heather is chatting, then so are Abbey and Kevin.

Explain your reasoning.

Exercise 6. (Rosen, exercise 36, sec. 1.2) The police have three suspects for the murder of Mr. Cooper: Mr. Smith, Mr. Jones, and Mr. Williams. Smith, Jones, and Williams each declare that they did not kill Cooper. Smith also states that Cooper was a friend of Jones and that Williams disliked him. Jones also states that he did not know Cooper and that he was out of town the day Cooper was killed. Williams also states that he saw both Smith and Jones with Cooper the day of the killing and that either Smith or Jones must have killed him. Can you determine who the murderer was if

1. one of the three men is guilty, the two innocent men are telling the truth, but the statements of the guilty man may or may not be true?
2. innocent men do not lie?

Exercise 7. The negation of the statement "if it rains, the ground is wet" is

- ◯ if the ground is not wet, it does not rain.
- ◯ it rains and the ground is not wet.
- ◯ if the ground is wet, it does not rain.
- ◯ if it rains, the ground is not wet.

Exercise 8. Let p and q be two propositions. Consider the two compound propositions below.

$$((q \to p) \land \neg q) \to \neg p \qquad\qquad \big(((\neg q) \to (\neg p)) \land p\big) \to q$$

- ◯ One of the compound propositions is a tautology, the other is a contingency.
- ◯ Both compound propositions are contingencies.
- ◯ One of the compound propositions is a contradiction, the other is a contingency.
- ◯ One of the compound propositions is a tautology, the other is a contradiction.

Exercise 9. The compound proposition $((\neg p \land q) \to (r \oplus q)) \lor (\neg s \leftrightarrow p)$ is

- ◯ a tautology.
- ◯ a contingency.
- ◯ a contradiction.
- ◯ incorrectly formatted.

Exercise 10. For each of the following propositions, determine if it is in CNF (Conjunctive Normal Form), DNF (Disjunctive Normal Form), both, or neither.

1. $(p \wedge q) \vee (\neg p \wedge \neg q)$
2. $(p \vee q) \wedge (\neg p \vee r)$
3. $(p \wedge q) \wedge (r \vee s)$
4. $(p \vee q) \vee (r \wedge s)$
5. $p \vee q$
6. $p \wedge q$

Exercise 11. Given the proposition: $p \to (q \wedge r)$, find:

1. The DNF (Disjunctive Normal Form) using a truth table.
2. The CNF (Conjunctive Normal Form) using a truth table.